

Cell populations at gastrulation

CONCEPTUS WITH A BILAMINAR EMBRYONIC DISC

At stage 6, the conceptus is composed of the walls of three cavities: the large chorionic cavity is surrounded by a meshwork of trophoblast and developing villi, and lined with extraembryonic mesoblast. The chorion, trophoblast and extraembryonic mesoblast enclose the extraembryonic coelom and contain the much smaller amniotic cavity and yolk sac (see Fig. 9.1). These latter cavities abut at the embryonic bilaminar disc where the epithelial epiblast and visceral hypoblast are approximated. A fourth cavity, the allantois, will form as a hypoblastic diverticulum in stage 7. The 'bilaminar disc' commonly referred to in embryology texts does not yet possess the definitive layers of embryonic ectoderm and endoderm that will give rise to embryonic structures. Only the epiblast will give rise to the embryo; all other layers produced so far are extraembryonic. The amnion and chorion (and surrounding mesoblast) are part of the extraembryonic somatopleure, whereas the yolk sac, allantois and surrounding extraembryonic mesoblast constitute extraembryonic splanchnopleure. At the junctional zone surrounding the margins of the embryonic area, where the walls of the amnion and yolk sac converge, the somatopleuric and splanchnopleuric layers of extraembryonic mesoblast are continuous.

The terms epiblast and hypoblast are used to make the distinction between the earliest bilaminar disc layers and the later embryonic layers. Epiblast and hypoblast contain mixed populations of cells with little restriction, which establish the placental structures and extraembryonic tissues before the production of embryonic cell lines at gastrulation. The older terminology depicting three germ layers that give rise to the skin, gut lining and intervening tissues is thus incorrect for the bilaminar and trilaminar embryonic disc. The application and retention of this aged terminology for the early stages of embryology continue to cause confusion and inhibit the development of more pertinent descriptive language to describe these early events. The earliest three cell lineages, not layers, that give rise to the extraembryonic membranes and the embryo are trophoblast, hypoblast and epiblast; each expresses different lineage-specific genes for its establishment, maintenance and differentiation (Artus and Hadjantonakis 2012).

At early stage 6, the epiblast is producing extraembryonic mesenchyme from its caudal margin. With the appearance of the primitive streak, a process is begun whereby cells of the epiblast either pass deep to the epiblast layer to form the populations of cells within the embryo, or remain on the dorsal aspect of the embryo to become the embryonic ectoderm. Although human embryos do not form a 'gastrula' as such, the term gastrulation is used here to denote an early period of development during which significant rearrangements, migrations and folding of the early embryo occur. The primitive streak is the site of organizer cells analogous to those found in embryos that do undergo gastrulation. The appearance of the primitive streak therefore marks the beginning of a period when gross alterations in morphology and complex rearrangements of cell populations occur. During this time, the epiblast will give rise to a complex multilaminar structure with a defined cranio-caudal axis. By the end of gastrulation, cell populations from different, often widely separated, regions of the embryonic disc will become spatially related and the embryonic shape will have been produced.

Primitive streak and node

Seen from the dorsal (epiblastic) aspect, at stage 6, the embryonic disc appears elongated. The primitive streak is first seen in the caudal region of the embryonic disc at this stage as a collection of pluripotent cells, orientated along its long axis in the median position, conferring the future cranio-caudal axis of the embryo (Figs 10.1–10.2). Although the future cranial and caudal regions of the embryo are well within the boundaries of the embryonic disc, it has become the practice to term

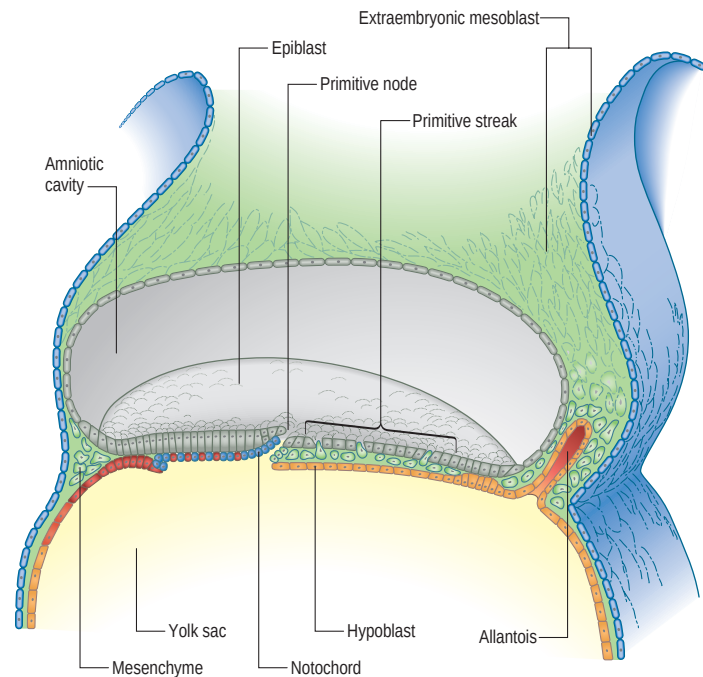


Fig. 10.1 A longitudinal section through an early conceptus. Ingression of mesoblast is occurring at the primitive streak and the notochord is ingressing via the primitive (Hensen's) node.

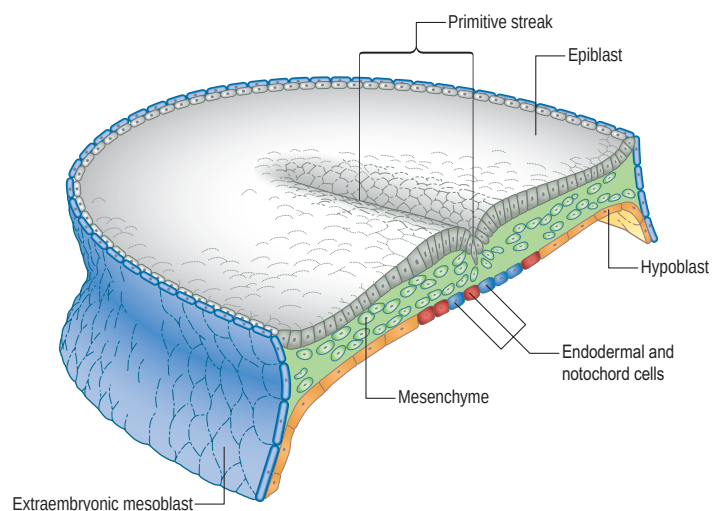


Fig. 10.2 A transverse section through the embryonic plate at the level of the primitive streak to show the early movement of mesoblast between the epiblast and underlying hypoblast.

the region of the disc closest to the streak 'caudal', and the region of the disc furthest from the streak 'cranial' or 'rostral'. With the development of the streak, the terms medial and lateral can be used. The relative dimensions of the primitive streak and the fates of the cells that pass through it change with the developmental stage. Thus, the streak extends half way along the disc in the stage 6 embryo, reaching its greatest relative length in stage 7 and its maximum length in stage 8.

Formation of the primitive streak is induced by the underlying visceral hypoblast, which remains beneath the streak even at later stages.

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The primitive streak may be considered to be generally homologous with the blastopore of lower vertebrates (e.g. amphibia), with the nodal region corresponding to the dorsal lip. Experiments clearly show the lip of the blastopore to be a dynamic wave front on which cells are carried into the interior to form the roof of the archenteron, a situation analogous to ingression through the node of the prechordal plate and endoderm. The primitive streak similarly may be considered analogous to the coapted, or fused, lateral lips of the blastopore, and the cloacal membrane and its immediate environs are considered analogous to the ventral lip of the blastopore.

At the primitive streak, epiblast cells undergo a period of intense proliferation, the rate of division being much faster than that of blastomeres during cleavage. Streak formation is associated with the local production of several cell layers, extensive disruption of the basal lamina, increase in adhesive plaques and gap junctions, synthesis of vimentin, and loss of cytokeratins by the emerging cells.

As the epiblast cells proliferate, two ridges are formed on each side of the primitive streak, which appears to sink between them. The lower midline portion of the streak is termed the primitive groove. The process by which cells become part of the streak and then migrate away from it beneath the epiblast is termed ingression.

The primitive node, or Hensen's node, is the most rostral region of the primitive streak. It appears as a curved ridge of cells similar in shape to the top of an old-fashioned keyhole. Cells ingressing from the ridge pass into the primitive pit (the most rostral part of the primitive groove), and then migrate rostrally beneath the epiblast. The primitive node has been recorded in all stage 7 human embryos; it produces axial cell populations, the prechordal plate, notochord, embryonic endoderm and the medial halves of the somites. Experimental removal of the node results in complete absence of the notochord and a failure of neurulation.

Recent studies on the dynamics of ingression at the primitive streak have shown movement of the underlying extracellular tissues as well as of the overlying cells. Re-examination of the processes of cell movement relative to the surrounding extracellular molecules in embryos has found that vertical and convergent extension motion patterns, previously thought to be limited to the epiblast, also occur in sub-epiblastic extracellular matrix fibrils of fibronectin (Zamir et al 2008, Szabó et al 2011).

Position and time of ingression through the primitive streak

Studies of cell fate have shown that epiblast cells that will pass through the streak are randomly located within the epiblast layer before their ingression, and that epiblast fate is determined at or before the time of ingression through the streak, indicating that passage through the primitive streak is the most important factor for future differentiation.

The position and time of ingression through either streak or node directly affect the developmental fate of cells. Passage through the streak is specified according to position, e.g. via the node, or rostral, middle or caudal regions of the streak. Cells that ingress through the primitive node give rise to the axial cell lines, the prechordal mesenchyme and notochord, and to the endoderm and the medial halves of the somites. The rostral portion of the primitive streak produces cells for the lateral halves of the somites, whereas the middle streak produces the lateral plate mesoblast. The adjacent caudal portion of the streak gives rise to the primordial germ cells, which can be distinguished histologically and histochemically, and the most caudal portion of the streak contributes cells to the extraembryonic mesoblast until the somites are visible. A composite of the information on the position of ingression through the streak and node is shown in Figure 10.3. The epiblast cells that do not pass through the streak but instead remain within the epiblast population give rise to the neural and surface ectoderm of the embryo.

Prechordal plate

The earliest cells migrating through the primitive node and streak give rise to both the embryonic endoderm and the notochord. The prechordal plate is first seen at stage 7. It has been defined as a localized thickening of the endoderm rostral to the notochordal process, although it is seen as a highly developed mesenchymal mass in contact with the floor of the neural groove, rostral to the notochordal process, rather than as an epithelial layer. The prechordal plate is a temporary collection of cells that underlies the neural plate during stage 9. It is composed of cells that are similar to, or larger and more spherical than, the ingressing endodermal cells (Müller and O'Rahilly 2003). In stage 8 embryos, the prechordal plate is up to eight cells deep and extends

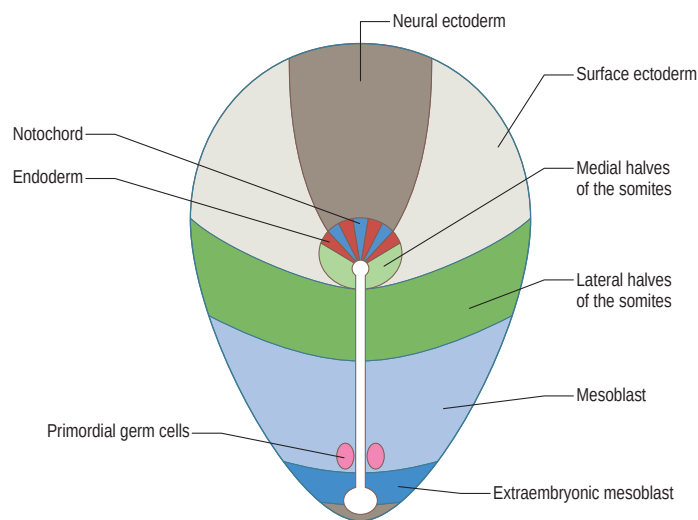


Fig. 10.3 The predictive fates of the epiblast cell population at the time the primitive streak is present.

along the long axis of the embryonic disc. By stages 9 and 10, the cells at the lateral edge of the plate have begun to migrate laterally as free mesenchymal cells and the plate reduces in height to two cells deep. At stage 11, the migrating prechordal mesenchyme forms bilateral pre-mandibular mesenchymal condensations and is no longer a median structure. The extent of prechordal cells remaining within the endoderm is not clear.

Notochord

The notochord, also called chordamesoderm, the head process or chorda, arises from epiblast cells of the medial part of the primitive node. It passes through several stages during development. The cells of the early notochordal process express myogenic markers transiently as they migrate beneath the epiblast, but later they become epithelial, forming junctions and a basal lamina. The notochordal cells are intimately mixed with endodermal cells, as both cell lines ingress at the same time (see Figs 10.1–10.2; Fig. 10.4). In the stage 8 embryo, the ingressing notochordal cells remain in the midline along the cephalocaudal axis. They form a rostral part, which is composed of a cell mass continuous with the prechordal mesenchyme; a mid portion, in which cells are arranged in a tube with a central notochordal canal; and a caudal epithelial layer of cells, the notochordal plate, which is contiguous with the embryonic endoderm and forms a roof to the secondary yolk sac. There is a transitory opening between the primitive node (and amniotic cavity) and the secondary yolk sac called the neurenteric canal (so named because its upper opening is in the future caudal floor of the neural groove, and its lower opening is into the archenteron, which is the primitive gut); it may still be found at stage 9, and the site of the neurenteric canal can be recognized in stage 10 embryos. The ingression of notochordal cells at the primitive node is matched by specification of the overlying neural ectodermal cells, and the notochordal plate is thus matched in length by the future neural floor plate. Both the notochord and the region of the floor plate of the neural tube may arise from a common progenitor cell. The early notochord is important for the maintenance and subsequent development of the neural floor plate and the induction of motor neurones. Removal of the notochord results in elimination of the neural floor plate and motor neurones, and expression of sensory cell types.

Caudal eminence

From stages 9 and 10, the region between the neurenteric canal and the cloacal membrane (see below), including the primitive streak, is termed the caudal eminence. It consists of the caudal region of the trunk, composed of mesenchyme derived from the primitive streak and epiblast, and covered with surface ectoderm. Whereas ingression of cells through the primitive streak gives rise to the prechordal and notochordal plates, and cells rostral to the neurenteric canal (see below), the cells of the caudal eminence arise from local division of a mesenchymal population positioned caudal to the neurenteric canal. The caudal portions of the notochord, which form later in development when secondary neurulation processes begin, arise from these cell populations, sometimes termed the caudoneural hinge or junction. This tissue is thicker and

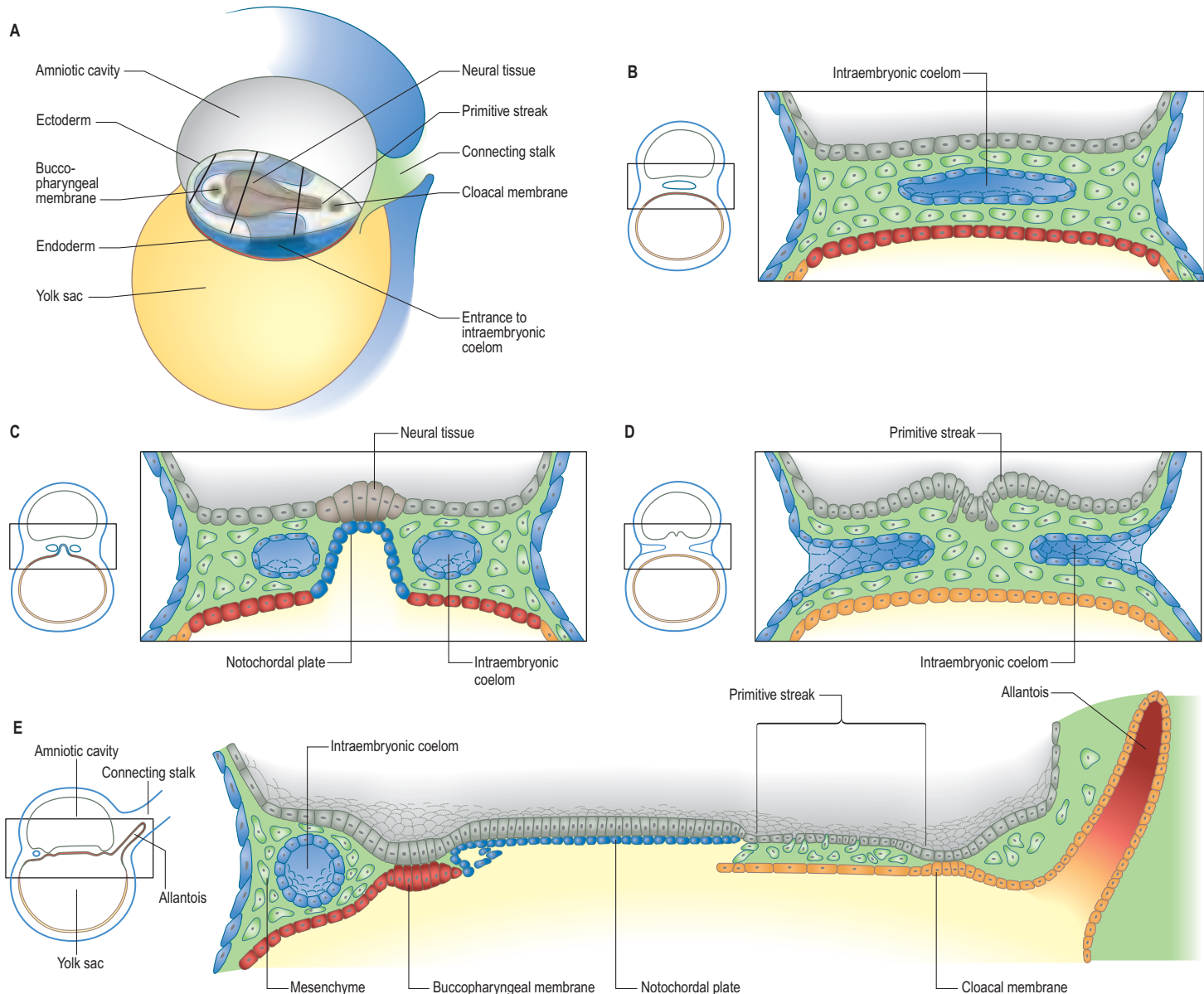


Fig. 10.4 **A**, The unfolded embryo, showing the disposition of the intraembryonic coelom within the embryonic disc. The lines across the embryo show the level of transverse sections through the disc shown in **B–D**. **B–D**, Transverse sections through the disc at the points indicated in **A**. **E**, A longitudinal section through the disc.

more advanced in differentiation than the tissues derived from the early primitive streak.

Embryonic endoderm

Before ingress, definitive embryonic endoderm cells are found in the epiblast, located at the primitive node and rostral primitive streak. In the mouse, the endodermal cells lie beneath the epiblast mainly in the midline, interspersed with presumptive notochordal cells, forming the roof of the secondary yolk sac. The ingressing endoderm displaces the visceral hypoblast into the secondary yolk sac wall by a dramatic territorial expansion that is brought about by a change in the morphology of the cells (see Figs 10.1–10.2, 10.4). The putative endoderm cells are cuboidal epithelial cells within the node but they become squamous in the endoderm layer; this could result in a four-fold increase in the surface area covered by the cells. A complete replacement of the visceral hypoblast has not yet been confirmed and there may be a mixed population of cells in the endodermal layer in the early stages. Ingression of cells through the streak and node in the human is apparent at stage 6, and, by stage 7, a population of endoderm and notochord cells is present beneath the epiblast (see Figs 10.1–10.2, 10.4). During stages 6–11, the midline roof of the secondary yolk sac becomes populated mainly by the notochordal plate, which remains in direct lateral continuity with the endodermal cells. It is not until stage 11, after the definitive notochord is formed, that the endoderm cells can join across the midline. For the developmental fate of the embryonic endoderm, see Figure 12.3.

Intraembryonic mesoblast (mesenchyme)

Epiblast cells ingress through the cranial and middle parts of the streak individually, maintaining their apical epithelial contacts while elongating ventrally. The cells become flask-shaped, with thin, attenuated apical necks and broad basal regions. Their basal and lateral surfaces form lamellipodia and filopodia, and the apical contact is released. The cells are now free mesoblast cells, their fibroblastic, stellate morphology reflecting the release from the epithelial layer. Once through the streak, the cells migrate away from it, using the basal lamina of the overlying epiblast and extracellular matrix as a substratum. The cells contact one another by filopodia and lamellipodia, with which they also contact the basal lamina. Gap junctions have been observed between filopodia and cell bodies. With the appearance of the mesoblast, spaces form between the epiblast and visceral hypoblast that are filled with extracellular matrix rich in glycosaminoglycans. The migrating mesoblast has a leading edge of cells that open up the migration routes, and the following cells seem to be pulled along behind in a coordinated mass movement. Mesoblast formed by cells migrating through the primitive node and rostral primitive streak will form the paraxial mesenchyme, whereas cells migrating through the middle to caudal streak will form the lateral plate mesenchyme (see Figs 10.1–10.2).

Embryonic ectoderm

When the ingress of cells through the primitive streak is completed, the epithelial cells remaining in the epiblast layer are termed embryonic

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ectoderm cells. This layer still contains a mixed population because both surface ectoderm cells and neural ectoderm cells are present. It is believed that these cells were originally in the cranial half of the disc when the primitive streak first appeared, at which time the neural-fated cells were closest to the streak, and the surface ectoderm cells were most cranial (see Fig. 10.3). The process of primary neurulation relocates most of the neuroepithelial cells (see below).

Primordial germ cells

Although early studies on human embryos have reported primordial germ cells, and described their development from the early endoderm of the yolk sac and allantois, it is now clear from animal experimentation that the primordial germ cells arise from epiblast ingressing at the caudal end of the primitive streak (see Fig. 10.3). It is not known whether these cells originate from rostral regions that migrate to the streak or from local caudal regions. Extremely early segregation of the germ cells, when the epiblast layer consists of only 10–13 cells, has been demonstrated. It has been suggested that the primordial germ cells remain sequestered in the extraembryonic mesenchyme at the caudal end of the embryo until the embryonic endoderm has been produced and gastrulation completed, and that they start to migrate along the allantoic and hindgut endoderm as the folding of the embryo begins. The formation of the tail fold brings the proximal portion of the allantois within the body, so reducing the final distance over which the cells migrate to the genital ridges. Further development of the germ cells is described in Chapter 72.

TRILAMINAR DISC

Although the stage 8 embryo is termed a trilaminar disc, the concept of three epithelial layers forming a trilaminar disc is incorrect; the middle, mesoblast, layer is several cells thick with intervening extracellular matrix. The embryo at this stage, approximately 23 days after ovulation, is pear-shaped, and broader cranially than caudally (Fig. 10.5). The upper epiblast cells are tall and form a pseudostratified columnar epithelial layer with a basal lamina, except at the primitive streak, where the cells are ingressing to form the other layers. The more centrally placed epiblast will give rise to neural ectoderm (neurectoderm) and the more laterally placed epiblast will give rise to surface ectoderm. The future neural ectoderm is seen as a neural plate that matches the length of the notochordal plate directly beneath, being slightly wider near the prechordal plate. The lower embryonic endoderm, a simple squamous layer with a developing basal lamina, is not always complete at this stage, particularly in the midline caudal to the prechordal plate, which is still occupied by the notochordal process or plate.

The middle, mesoblast, layer is composed of free cells migrating cranially, laterally and caudally from the primitive streak (see Fig. 10.4). They produce extracellular matrix, which separates the epiblast and endoderm of the embryonic area and permits their passage. The streams of mesoblast extend between the epiblast and endoderm over all of the disc area except cranially at the buccopharyngeal membrane (where the endoderm and ectoderm become apposed once the prechordal mesenchyme has migrated laterally), and caudally at the cloacal membrane (a patch of thickened endoderm, similar to the buccopharyngeal membrane, caudal to the primitive streak). The mesoblast on each side of the notochord is termed paraxial mesenchyme.

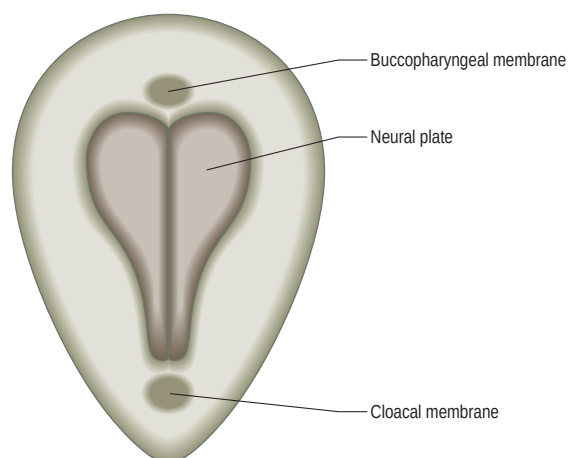


Fig. 10.5 The extent and shape of the neural plate in an unfolded embryo.

Mesoblast that passes in a cranial direction flanks the notochordal plate and passes around the prechordal plate region, converging medially to fuse in the midline beyond its cephalic border. This transmedian mass, in which the heart and pericardium will develop, is initially termed the cardiogenic mesoblast. It fuses with the junctional zone of extraembryonic mesoblast around the extreme cephalic margin of the embryonic area. This region will eventually form the septum transversum and primitive ventral mesentery of the foregut. Mesoblast passing laterally from the streak soon approaches and becomes confluent with the extraembryonic mesoblast around the margins of the disc, i.e. at the junctional zone where the splanchnic and somatic strata of extraembryonic mesoblast merge. Mesoblast that streams caudally from the primitive streak skirts the margins of the cloacal membrane and then converges towards the caudal midline extremity of the embryonic disc to become continuous with the extraembryonic mesoblast of the connecting stalk. It is unclear if the lower layer of the cloacal membrane consists of visceral hypoblast, like the more cranial primitive streak (the hypoblast is necessary for maintaining the streak), or if it is replaced by migrating embryonic endoderm, or if there is a region for ingression of endoderm at the caudal end of the streak, similar to the node cranially.

Still further caudally, the embryonic disc develops a midline diverticulum adjacent to the cloacal membrane. This diverticulum, the allantois, projects into the extraembryonic connecting stalk (see Figs 9.9, 10.4; Fig. 10.6). There is little information about which cells form the allantois, i.e. whether it is composed of visceral hypoblast, parietal hypoblast or embryonic endoderm. The allantois later develops a rich anastomotic blood supply around it, in the manner of the yolk sac.

The generation of cells at the primitive node produces midline endoderm, notochord and the floor plate of the future neural tube. As the notochord grows and elongates, there is a matched growth of neural floor plate cells until both cell lines extend to the buccopharyngeal membrane. The epiblast lateral to the midline contains both future surface and neural ectoderm. The latter becomes arranged between the primitive node and buccopharyngeal membrane; cells destined to be in the neural plate lie medially, and those destined for the neural crest lie along the junction between the neural plate and surface ectoderm (see Fig. 10.5). A smaller subpopulation of neuronal cells, the ectodermal placodes, are arranged either close to the neural crest or within the rostral limit of the neural plate itself.

FOLDING OF THE EMBRYO

In a diagrammatic representation of the trilaminar disc prior to folding and viewed from the ectodermal aspect, all of the future external surface of the body is delimited (see Fig. 10.5). The ends of the gut tube are specified on the ectodermal surface at the buccopharyngeal and cloacal membranes, which are regions where the ectoderm and underlying endoderm are apposed without intervening mesoblast. In the midline between these membranes, proliferation of the neural ectoderm matches the underlying migration of mesoblast from the primitive streak, so that the neural plate covers the paraxial mesenchyme on each side of the notochord (see Fig. 10.6E). As the paraxial mesenchyme segments, the formation of the epithelial somites elevates the edges of the neural plate and initiates primary neurulation (see Fig. 10.6F; Figs 10.7–10.8). The neural plate itself undergoes concurrent morphological changes. The most medial cells become wedge-shaped, forming the neural groove. Further elevation of the edges of the neural groove permits fusion of the neuronal populations in the dorsal midline to form the neural tube. The surface ectoderm forms the putative dorsal epidermis (see Figs 10.6G, 10.7–10.8). Cells at the lateral edge of the neural plate, termed neural crest cells, remain as a linearly arranged mesenchymal population between these two epithelia. Fusion of the neural tube begins in the future rhombencephalic region of the embryo and proceeds rostrally and caudally to about the level of somite 29. Neurulation is described further in Chapter 17. A population of neural epithelial cells remain within the surface ectoderm; at this stage they are termed ectodermal placodes.

The representation of a person on the trilaminar disc (Fig. 10.9) shows, to some extent, the way in which the positions of the main body structures are already specified in the unfolded embryo. Ectoderm lateral to the neural plate and the paraxial mesenchyme will form structures within the back. The portion of the disc between the buccopharyngeal membrane and the edge of the disc will become the ventral thoracic wall and the ventral abdominal wall cranial to the umbilicus. Further caudally, midway along the neural axis, the lateral portions of the disc will become the lateral and ventral abdominal walls of the

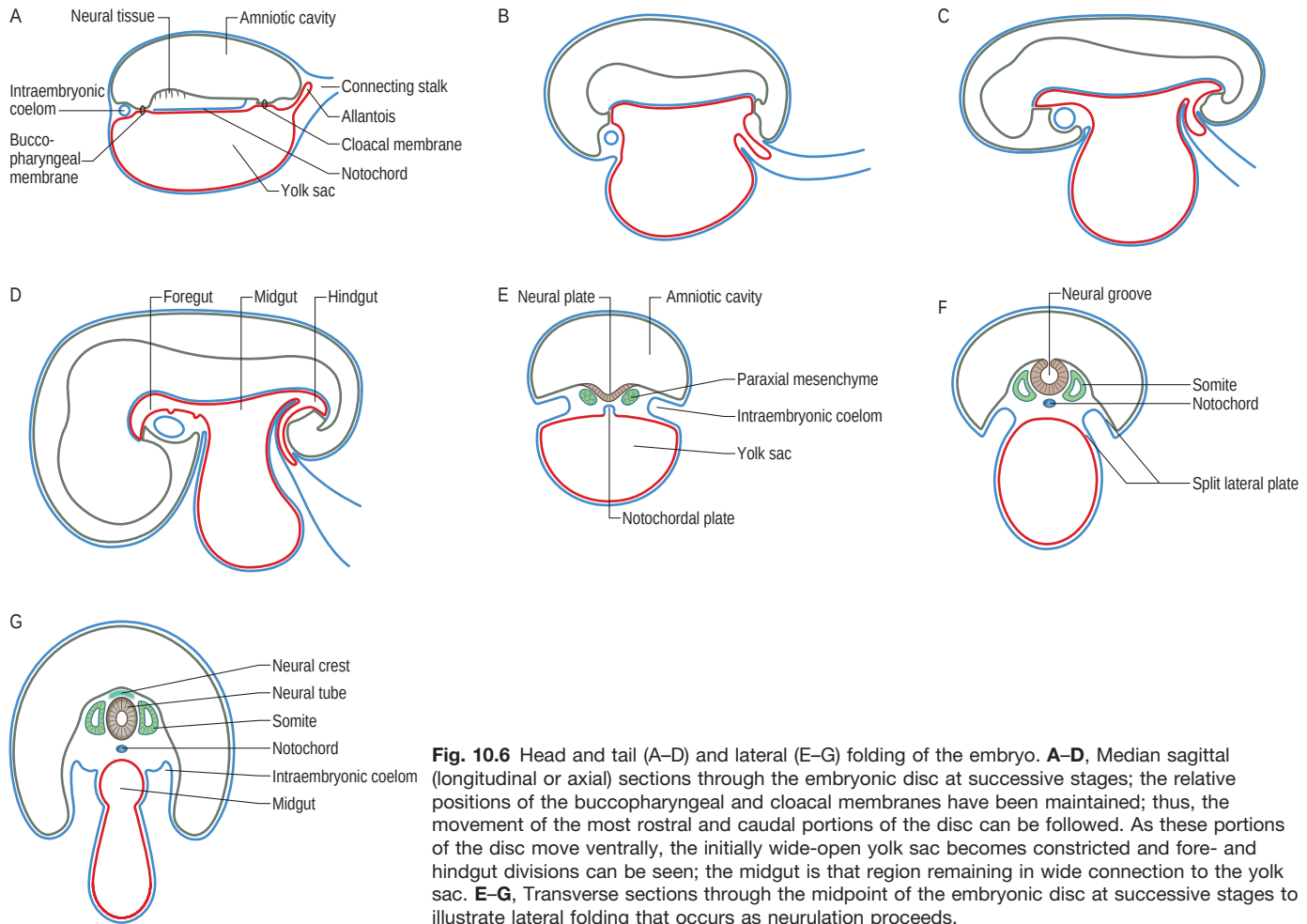


Fig. 10.6 Head and tail (A–D) and lateral (E–G) folding of the embryo. **A–D**, Median sagittal (longitudinal or axial) sections through the embryonic disc at successive stages; the relative positions of the buccopharyngeal and cloacal membranes have been maintained; thus, the movement of the most rostral and caudal portions of the disc can be followed. As these portions of the disc move ventrally, the initially wide-open yolk sac becomes constricted and fore- and hindgut divisions can be seen; the midgut is that region remaining in wide connection to the yolk sac. **E–G**, Transverse sections through the midpoint of the embryonic disc at successive stages to illustrate lateral folding that occurs as neurulation proceeds.

trunk. The portion of the disc beyond the cloacal membrane will form the ventral abdominal wall caudal to the umbilicus. The circumference of the disc, where the embryonic tissue meets the extraembryonic membranes, will become restricted to the connection between the ventral abdominal wall and the umbilical cord, i.e. the umbilicus.

Head folding begins at stage 9, when the fusing cranial neural plate rises above the surface ectoderm and the portion of the disc rostral to the buccopharyngeal membrane (which contains the cardiogenic mesenchyme) moves to lie ventral to the developing brain (see Fig. 10.6). The prosencephalon and buccopharyngeal membrane are now the most rostral structures of the embryo. The previously flat region of endoderm, which may contain cells from the prechordal plate, is now modified into a deep tube, the primitive foregut. Tail folding can be seen in stage 10 embryos, when the entire embryo comes to rise above the level of the yolk sac. Similar movement of the part of the disc caudal to the cloacal membrane results in its repositioning ventral to the neural plate. Generally, as the embryo rises above the edges of the disc, the lateral regions of the disc are drawn ventrally and medially, contributing to the lateral folding of the embryo.

FORMATION OF THE INTRAEMBRYONIC COELOM

At and just before stage 9 (before formation of the head fold), vesicles appear between the mesenchymal cells cranial to the buccopharyngeal membrane and within the cranial lateral plate mesenchyme. At the periphery of the vesicles, the mesenchymal cells develop junctional complexes and apical polarity, and form an epithelium. The vesicles become confluent to form a horseshoe-shaped tube, the intraembryonic coelom, which extends caudally to the level of the first somite and laterally into the lateral plate mesenchyme towards the extraembryonic mesenchyme. The lateral plate mesenchyme thus develops somatopleuric coelomic epithelium subjacent to the ectoderm, and a splanchnopleuric coelomic epithelium next to the embryonic endoderm (see Fig. 12.2C(iv)). At this stage, the intra- and extraembryonic coeloms do not communicate.

During development of the head fold, the morphological movements that organize the foregut and buccopharyngeal membrane have a similarly profound effect on the shape of the intraembryonic coelom.

The midline portion of the originally flat, horseshoe-shaped coelom moves ventrally, leaving the caudal arms of the horseshoe in their original position. In this way, the midline part of the coelom, which was originally just rostral to the buccopharyngeal membrane, comes to lie ventral to the foregut (caudal to the buccopharyngeal membrane), and the two lateral extensions of the coelom pass close to the lateral walls of the foregut on each side. The caudal portions of the coelom (the two arms of the horseshoe), which, in the unfolded disc, communicated laterally with the extraembryonic coelom, turn 90° to lie lateral to the gut, and communicate with the extraembryonic coelom ventrally.

Compartments of the coelom that will give rise to the body cavities later in development can already be seen. The midline ventral portion, caudal to the buccopharyngeal membrane, becomes the pericardial cavity. The canals lateral to the foregut (pericardioperitoneal canals) become the pleural cavities and the uppermost part of the peritoneal cavity. The remaining portion of the coelom becomes the peritoneal cavity. By stage 11, the intraembryonic coelom within the lateral plate mesenchyme extends caudally to the level of the caudal wall of the yolk sac. The intra- and extraembryonic coeloms communicate widely on each side of the midgut along the length of the embryo from the level of the fourth somite (Fig. 10.10).

In the early embryo, the intraembryonic coelom provides a route for the circulation of coelomic fluid and, with the beating of the heart tube, functions as a primitive circulation that takes nutritive fluid deep into the embryo until it is superseded by the blood vascular system. The coelomic channel, and the primitive circulation that passes through it, is of paramount importance up to stage 13. Whereas the superficial tissues of the embryo can receive nutrients via the amniotic sac and yolk sac fluids, the deeper tissues are, until the formation of the coelom, under conditions similar to those found in tissue culture. However, from stage 10, exocoelomic fluid, propelled by the first contractions of the developing heart, is brought into contact with the deeply placed mesenchyme. This early 'circulation' ensures that an adequate supply of nutrients reaches the rapidly increasing amount of embryonic tissue, and meets most of the requirements of the deeper mesenchymal derivatives. From stage 12, the endothelial system expands and fills rapidly with plasma, which passes across the locally thinned coelomic epithelium into the large hepatocardiac channels that project into the pericardioperitoneal canals at the level of the seventh somite.

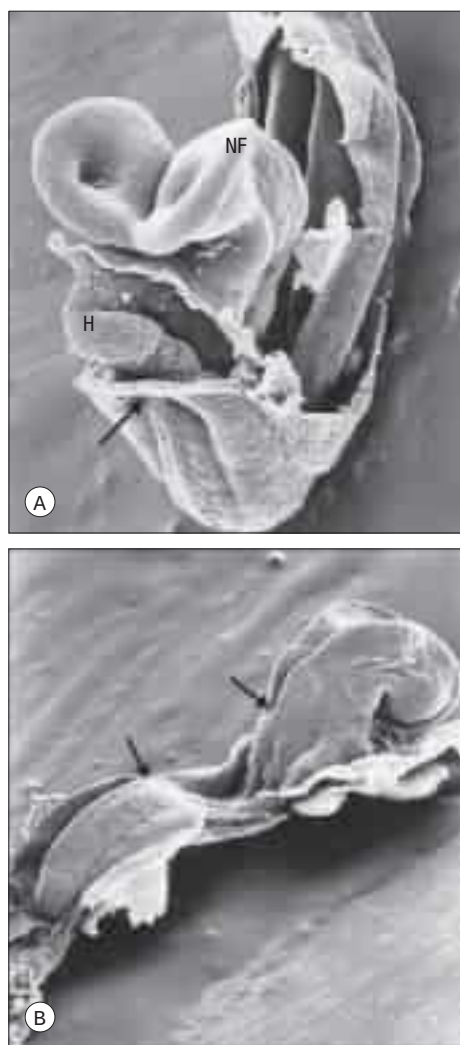


Fig. 10.7 Scanning electron micrographs of rat embryos at the time of neurulation. **A**, Ventral view, showing the neural fold (NF), and the heart (H), with the somatopleuric pericardial membrane and surface ectoderm removed; the arrow indicates the entrance to the foregut via the cranial intestinal portal. **B**, Dorsolateral view; the arrows indicate the extent of rostral (to the right) and caudal (to the left) neural tube formation. (Photographs by P Collins; printed by S Cox, Electron Microscopy Unit, Southampton General Hospital.)

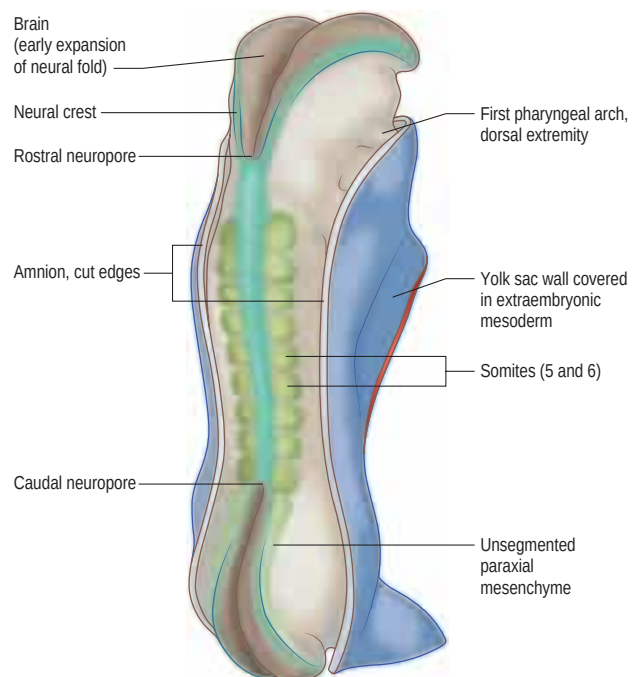


Fig. 10.8 A human embryo at stage 10, 2.1 mm long, with nine somites: right lateral and dorsal aspects. Nearly all the yolk sac and the caudal amnion have been excised.

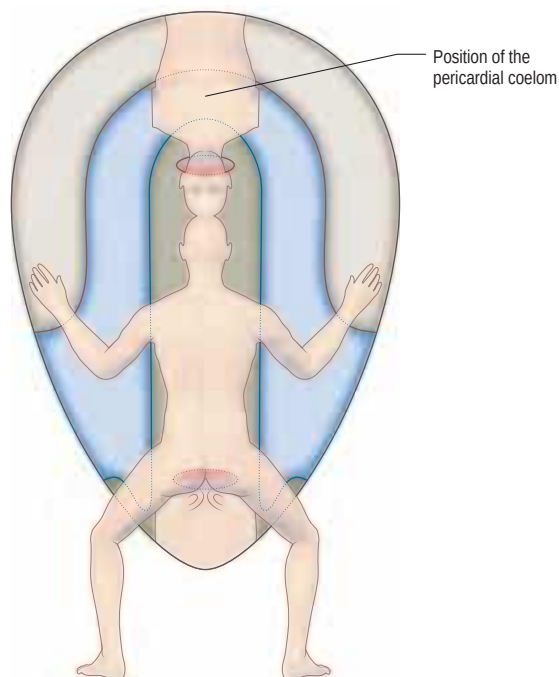


Fig. 10.9 A representation of a person on the flat embryonic disc. The position of the central nervous system has been matched to the dimensions of the neural plate, and the position of the heart in the thorax to the position of the pericardial coelom. The limbs, although represented in this diagram, are not present on the disc at this stage. The usefulness of this diagram lies in its illustration of the extent of the anterior body wall both rostral to the buccopharyngeal membrane and caudal to the cloacal membrane. The future dorsal regions of the body are found medially on the disc, while the ventral regions of the body are situated laterally and peripherally on the disc. After head and tail folding and lateral folding, the peripheral edge of the disc becomes constricted as the edge of the umbilicus.

In spite of the importance of the coelom in defining the body cavities, and of the coelomic epithelium in the production of the major mesenchymal populations of the trunk (Streeter 1942, Langemeijer 1976, O'Rahilly and Müller 1987) (see Fig. 12.2), until recently the overall contribution of the coelom and its epithelium to the embryo has received relatively little attention (Carmona et al 2013). The coelom is a single, tubular organ that may be compared to the neural tube, in that it possesses a specialized wall that encloses a cavity. Like neural ectoderm, proliferating coelomic epithelium is a pseudostratified columnar epithelium with an inner germinal layer, from which cellular progeny migrate. After the germinal phase, both epithelia ultimately form the lining of a cavity, ependyma for the neural epithelium, and mesothelium for the coelomic epithelium. Coelomic epithelium, like neural epithelium, produces cells destined for different fates from different sites and at different developmental times. Coelomic cells are like the neural epithelium, in that they have apical epithelial specializations and tapering basal processes that are in direct contact with the underlying mesenchyme, without an intervening basal lamina. The possibility of the tapering processes forming directional signals for migrating progeny, similar to the radial glia of the neural tube, has not been examined.

EMBRYONIC CELL POPULATIONS AT GASTRULATION

After gastrulation, the cells of the embryo contribute to two fundamental types of tissue, namely: epithelial and mesenchymal. Differentiation of specialized circulating blood cells and other cell types occurs in sequence. Embryonic and fetal cell types are replaced later in development or after birth.

Epithelia

Epithelial populations in the embryo have many of the morphological characteristics of differentiated epithelia, i.e. they are composed of sheets of closely packed, polarized cells, with narrow intercellular clefts containing minimal extracellular material, and a developed basal

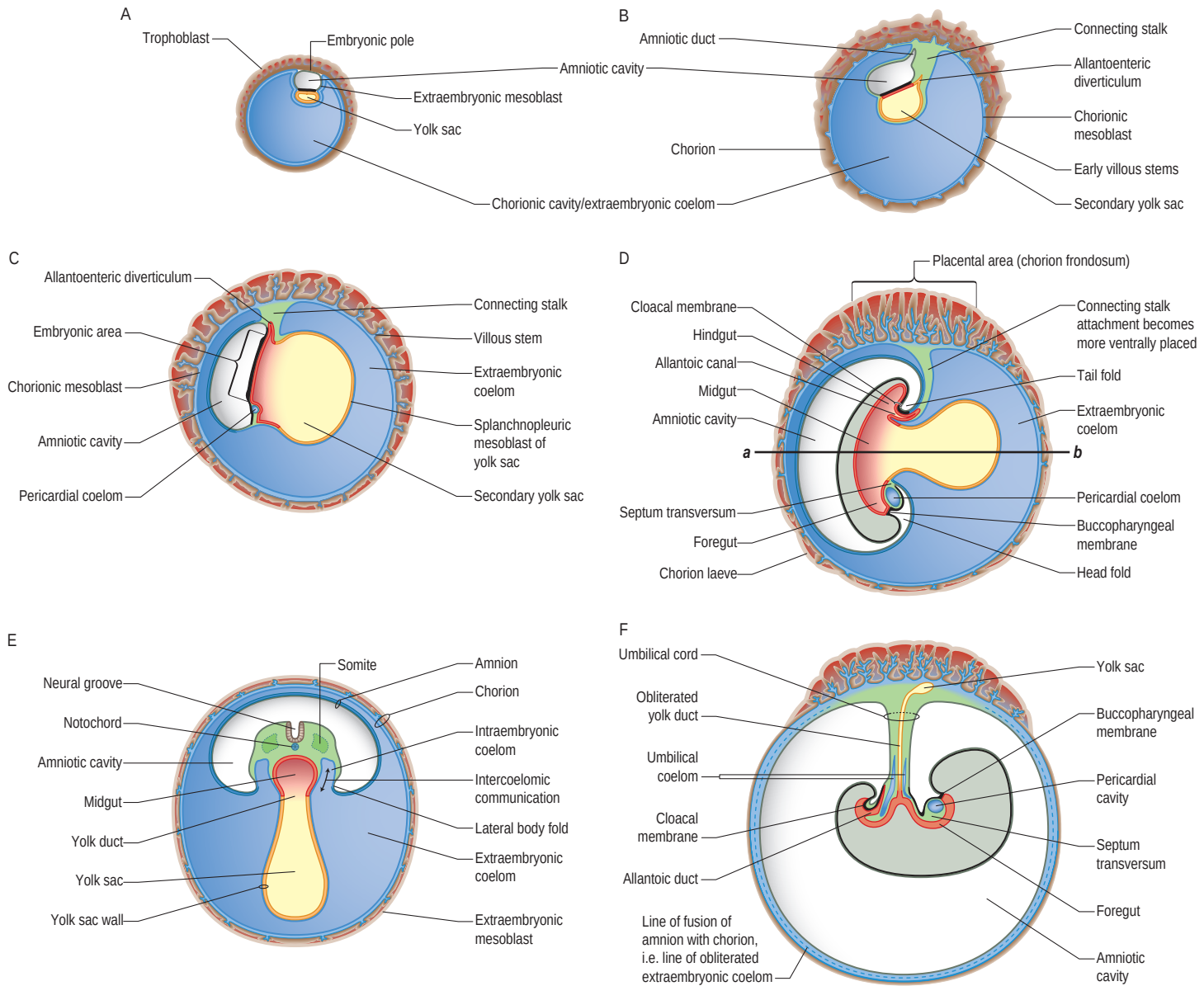


Fig. 10.10 **A**, An early stage in development of a human blastocyst. **B**, A blastocyst sectioned through the longitudinal axis of an embryo, showing the early formation of the allantois and the connecting stalk. **C**, A longitudinal section of an embryo at a later stage of development; the pericardial cavity can be seen at the most rostral part of the embryonic area. **D**, A longitudinal section of an embryo at a later stage, showing formation of the head and tail folds, the expansion of the amnion and the delimitation of the umbilicus. **E**, A transverse section along the line a–b in **D**; observe that the intraembryonic coelom communicates freely with the extraembryonic coelom. **F**, A longitudinal section of an embryo at a later stage, showing full expansion of the amniotic cavity and the umbilical cord.

lamina containing specific proteins synthesized by the epithelium itself. The cells usually show juxtaluminal lateral surface specializations, such as desmosomes, tight junctions, gap junctions, and so on, and specializations of the apical surface, such as microvilli or cilia.

Epithelial polarity factors establish the extent of the apical, lateral and basal domains of the individual cells forming an epithelium. Three main control pathways have been identified in maintaining apicobasal polarity (St Johnston and Sanson 2011). The apical domain is specified by the Crumbs transmembrane protein; Baz/PAR-3 specifies the position and extent of the intercellular junction; and Scribble restricts the size of the junctional domains. Once established with an underlying basal lamina, the basal surface displays integrins. Changes in the morphology of an epithelial sheet will involve relative changes in the apical, lateral and basal domains of all cells. Planar polarity of epithelia is driven by intercellular cortical actin and myosin. Epithelial cells are contiguous via actomyosin contractile networks, which contact the lateral cell membrane through E-cadherin adhesion (Gorfinkel and Blanshard 2011). Wnt/planar cell polarity signalling pathways are involved in these epithelial morphogenetic movements and also in orientated cell division which occurs in epithelial sheets (Tao et al 2009).

Embryonic epithelia differ from those in the fetus and adult. Two distinct types can be identified. Early germinal epithelia, which give rise to epithelial or mesenchymal populations of the embryo and confer

their early patterning are termed organizers, e.g. the primitive streak. Later germinal epithelia give rise to system-specific progenitor populations, e.g. the ventricular zone of the neural tube. All epithelia other than special germinal epithelia divide to produce embryonic growth throughout development and may retain stem cells that will divide throughout life.

Characteristically, epithelia clothe internal and external surfaces as simple or compound cellular sheets that separate phases of differing composition (e.g. the external environment and the subepithelial tissue fluids; intravascular and extravascular fluids, and so on). Traffic of materials in the intercellular clefts is limited. Traffic occurs across the cells because their limiting membranes, which function as energy-dependent selective barriers, enhance the passage of some materials and impede the passage of others.

Mesenchyme

The terms mesoblast and mesenchyme are used in this text in a specific manner and are not interchangeable. Previously, cells forming a population between the epiblast and hypoblast were termed mesoderm and, more recently, mesenchyme. The terms primary and secondary mesenchyme have been used to distinguish between those cells that arise from ingression through the primitive streak and those that arise from

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neural crest ingression, respectively. Primary mesenchymal cells revert to epithelia at their destinations. However, whereas some primary mesenchymal cells may become epithelial within a short timeframe, e.g. somites and lateral plate, other cells may transform later, e.g. the epithelium lining blood vessels. To address these terminological conflicts, the mixed population of epiblast cells that ingress through the primitive streak and come to lie between the epiblast and embryonic endoderm is termed mesoblast until the cells have migrated to their final position, at which time the populations of mesenchyme can be identified and their fates inferred.

Mesoblastic and mesenchymal cells have no polarity. They form junctional complexes, which are not exclusively juxtaluminal, and they produce extracellular matrix molecules and fibres from the entire cell surface. Mesenchymal populations are formed from a range of germinal epithelia and by proliferation of mesenchymal cells directly, and occupy all the regions between the various epithelial layers described above. The term mesoderm is reserved for the coelomic epithelia that later form mesothelia.

Mesenchymal cells support epithelia throughout the developing body, both locally where they contribute to the basement membrane and form the lamina propria and smooth muscle of tubes, and generally where they differentiate into connective tissue. Specific mesenchymal populations control the patterning of local regions of epithelium (e.g. the zone of polarizing activity on the postaxial limb border posterior to the apical ectodermal ridge).

Extracellular matrix

The space beneath epithelia and between mesenchyme cells is filled with extracellular matrix; both epithelial and mesenchymal cells synthesize extracellular matrix molecules and their receptors. Epithelial cells produce a two-dimensional basal lamina, which contains a variety of matrix molecules, including laminin, fibronectin, type IV collagen and various proteoglycans. The particular molecules can vary during development according to spatial and temporal patterns, resulting in changes in the behaviour of the underlying mesenchymal cells (e.g. in patterning of the basal regions of the skull). Mesenchymal cells produce extracellular matrix molecules in three dimensions. Those adjacent to an epithelial layer will connect with its basal lamina, forming a basement membrane that secures the epithelial layer to the underlying tissue. Cells deep within a mesenchymal population may synthesize matrix molecules (fibrillar or granular) to separate cells locally, open up migration routes or leave information within the matrix to act on cell populations passing at a later time.

Molecules of the extracellular matrix are complex; they include more than 19 individual types of collagen (some of which are capable of being individually spliced to give more than 100 variants), proteoglycans and glycoproteins (which come in a wide variety of forms, with and without binding proteins), and elastic fibres (see Ch. 2). Hyaluronic acid, a glycosaminoglycan, has a vast capacity to bind water molecules and so create and structure the space between the mesenchymal cells, thereby producing much of the overall shape of an embryo. Experimental removal of hyaluronic acid prevents the formation of cell migration

routes, removes the support for overlying epithelia, and disrupts the normal branching of glandular systems.

Fibronectin deposited extracellularly along a migration pathway will affect cells that touch it later, causing realignment of their intracellular actin filaments and thus of their orientation; it also induces cell migration. The receptors for extracellular matrix molecules such as fibronectin and laminin were originally termed integrins because they integrate extracellular proteins and intracellular cytoskeletal elements (via α and β subunits that span the cell membrane), allowing them to act together; the binding preference of integrins depends on their combination of subunits and environmental conditions.

It is known that extracellular matrix is structured rather than random. Epithelial and mesenchymal cell populations can structure the space around them by secretion of particular matrix molecules or growth factors, which, in turn, can organize the cells that contact them. Cell–matrix interactions and matrix–cell interactions control the position of migration routes and cellular ‘decisions’ to migrate or to begin to differentiate. Matrix molecules propagate developmental instructions from cell to cell and form a far-reaching four-dimensional (spatial and temporal) mechanism of communication. Mapping the molecular pathways and the biomechanical processes that underlie morphogenesis, the dynamic epigenetics of early mesenchyme cells and their secreted three-dimensional extracellular matrices, and the mechanisms by which the stiffness of the surrounding matrix affects the actions of the cells within it and the specific deposition of fibrillary proteins creates tissue boundaries and zones of tension, are all the subject of ongoing research (e.g. Davidson et al 2010, Loganathan et al 2012).

Transition between epithelial and mesenchyme states

Transformations of cell morphology from epithelium to mesenchyme, and *vice versa*, occur in specific places and times during development, and can be seen as ways of dispersing germinal centres with increasing restriction. The first epithelial-to-mesenchyme transition occurs at the primitive streak, a germinal epithelium that confers embryonic specification on the resultant mesoblast population. The mesoblast so formed migrates and the cells undergo mesenchyme-to-epithelial transitions when they reach their final destinations. Series of small epithelial germinal centres, the somites, are formed, as are larger, more extensive, germinal epithelial sheets that line the walls of the intraembryonic coelom. The coelomic walls, especially those derived from somatopleure and splanchnopleure, form germinal epithelia that give rise to the major mesenchymal populations that form the viscera. The early epithelial somites undergo further local epithelial-to-mesenchyme transitions to form the sclerotomes, and subsequently form several germinal epithelia in the epithelial plate of each somite. Later mesenchyme-to-epithelium transitions are not associated with the formation of germinal epithelia; the most common involve the transition of mesenchyme into the endothelium of the vascular system. The nephrons of the mesonephric and metanephric systems also form from mesenchyme-to-epithelial transition.

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